Sri Lanka Institute of Information Technology

IE2044 – Systems Modelling and Prototyping

System Modeling and Prototyping of an LM35/ICL7107-Based Digital Thermometer

(Group 15)

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Introduction

This project aims to design and build a 2-digit digital thermometer capable of measuring temperatures from 0°C to 99°C. The system uses an LM35 precision temperature sensor and an ICL7107 analog-to-digital converter (ADC) with integrated 7-segment display drivers.

The LM35 produces an analog output voltage linearly proportional to temperature, with a scale factor of 10 mV per degree Celsius. For example, at 25°C the output is 250 mV, and at 50°C it is 500 mV. The ICL7107 functions as a digital voltmeter chip, converting an analog input voltage directly into numerical readings on standard 7-segment displays without requiring a microcontroller.

The key design challenge is voltage scaling. The ICL7107 requires an external reference voltage (Vref) to interpret the analog input. By setting Vref to exactly 100 mV using a potentiometer, the chip is configured so that every 10 mV from the LM35 is displayed as the digit "1". Consequently, a 250 mV input (25°C) should display "25", a 500 mV input (50°C) should display "50", and so on up to 99°C.

The project was executed in three sequential phases:

1. **Proteus Simulation** – To verify voltage scaling, reference adjustment, and overall functional behavior of the LM35 and ICL7107 combination before committing to hardware.
2. **EasyEDA PCB Design** – To create a manufacturable printed circuit board following best practices, including analog star grounding to prevent display noise and horizontal alignment of the two 7-segment displays.
3. **Breadboard Prototyping** – To physically implement and test the circuit as an intermediate step before final PCB assembly and soldering.

Project Outcome Summary:

Phase	Status	Remarks
Proteus Simulation	Successful	Accurate temperature display from 0°C to 99°C with correct Vref adjustment.
EasyEDA PCB Layout	Completed	Gerber files generated. Star grounding and display alignment implemented.
Breadboard Prototype	Not Functional	Display consistently showed **"66"** regardless of input voltage or temperature.

This report documents the successful simulation and PCB design work. It then analyzes the specific "66" failure mode observed during breadboard testing, identifies probable causes, and outlines a structured future correction plan to achieve a fully working hardware prototype. The experience highlights the critical gap between idealized simulation environments and real-world circuit implementation challenges, including oscillator requirements, power supply stability, decoupling, and proper grounding.

Design Methodology

The project was carried out in three distinct phases: Proteus simulation, EasyEDA PCB design, and breadboard prototyping. Each phase is described below.

Phase I: Proteus Simulation

The first step was to simulate the complete circuit in Proteus to verify functionality before any hardware work.

Steps taken:

1. **Component placement** – An LM35 sensor, ICL7107 ADC, two 7-segment displays (common anode), a potentiometer, and necessary passive components (resistors and capacitors) were placed in the Proteus workspace.
2. **Circuit connection** – The LM35 output was connected to the analog input (IN+) of the ICL7107. The ICL7107's digit driver outputs were connected to the two 7-segment displays. A potentiometer was connected to the reference voltage input (Vref) of the ICL7107.
3. **Reference voltage adjustment** – The potentiometer was adjusted until the reference voltage (Vref) of the ICL7107 measured exactly 100 mV. This ensures that a 10 mV input from the LM35 corresponds to the digit "1" on the display.
4. **Temperature testing** – The LM35 temperature was varied in simulation from 0°C to 99°C, and the displayed value on the two 7-segment displays was recorded and verified against the expected temperature.

Phase II: EasyEDA PCB Design

After successful simulation, a printed circuit board (PCB) was designed in EasyEDA Pro for future fabrication and assembly.

Steps taken:

1. **Component selection** – Through-hole LM35 (TO-92 package) was chosen so it could extend outside an enclosure. ICL7107 in DIP-40 package was selected for easy soldering. Resistors and capacitors were selected in SMD 0805 package for compact layout.
2. **Schematic capture** – The verified simulation schematic was recreated in EasyEDA Pro with correct component footprints.
3. **PCB layout** – Components were placed on the board with attention to the following constraints:
    - **Analog grounding (Star Point)** – The LM35 ground and the ICL7107 analog ground pins were connected together at a single "star point" to prevent digital noise from corrupting the temperature reading.
    - **Display alignment** – The two 7-segment displays were placed perfectly horizontally aligned for proper visual appearance.
    - **Power supply decoupling** – Capacitors were placed close to the ICL7107 power pins to reduce noise.
4. **Design Rule Check (DRC)** – The layout was verified for clearance and connectivity errors.

5. **Gerber generation** – Final manufacturing files (Gerbers) were exported for future fabrication.

Phase III: Breadboard Prototyping

After completing the PCB design, a breadboard prototype was constructed to test the circuit before committing to PCB fabrication.

Steps taken:

1. **Component placement** – The LM35, ICL7107 (DIP-40), two 7-segment displays, potentiometer, resistors, and capacitors were inserted into a breadboard following the same schematic used in simulation and PCB design.
2. **Power supply connection** – A +5V power supply was connected to the ICL7107. The negative supply (V-) pin was connected to ground (note: this was later identified as a potential issue, as the ICL7107 typically requires ±5V).
3. **Input testing** – The LM35 output was measured with a multimeter to confirm correct voltage output at room temperature (approximately 250 mV). An external DC voltage source was also used to inject known voltages (0 mV, 250 mV, 500 mV, 990 mV) directly into the ICL7107 input for isolation testing.
4. **Reference adjustment** – The potentiometer was adjusted while measuring Vref with a multimeter, attempting to achieve 100 mV.
5. **Observation** – The display behavior was recorded under all test conditions.

Component Selection

All components were selected based on the project requirements: temperature range of 0°C to 99°C, direct 7-segment display output, through-hole sensor for external mounting, and a mix of through-hole and SMD parts for PCB design. The complete bill of materials is presented below.

Bill of Materials

Component	Package	Quantity	Key Specifications	Selection Reason
LM35	TO-92 (through-hole)	1	10 mV/°C output, ±0.5°C accuracy	Precision analog temperature sensor; through-hole allows mounting outside enclosure
ICL7107	DIP-40	1	3.5 digit ADC with 7-segment drivers	Converts analog voltage directly to display; no microcontroller needed
7-segment display	Common anode	2	0.56 inch digit height	Standard display for 2-digit temperature readout
Potentiometer	10 kΩ (trimmer)	1	Multi-turn recommended	For precise adjustment of Vref to 100 mV
Resistors	SMD 0805	Various	Per ICL7107 datasheet	Current limiting for displays and reference network

Capacitors	SMD 0805	Various	100 nF, 47 pF, etc.	Power decoupling and oscillator circuit
Power supply	–	1	±5V DC	Required for proper ICL7107 operation

Justification of Key Components

LM35 (TO-92 through-hole)

- Provides linear 10 mV/°C output without external calibration.
- Through-hole package allows the sensor to be mounted on a short wire and placed outside the PCB enclosure for accurate ambient temperature sensing.
- Operating range (-55°C to +150°C) exceeds project requirements.

ICL7107 (DIP-40)

- Integrates ADC, reference voltage generator, and 7-segment drivers in a single chip.
- DIP-40 package is large but easy to solder on breadboard and PCB.
- Directly drives common anode 7-segment displays without additional driver ICs.

7-Segment Displays (Common Anode)

- Compatible with ICL7107 output architecture.
- Two displays provide 2-digit readout for 0–99°C range.
- 0.56 inch digit height ensures good visibility.

Potentiometer (10 kΩ trimmer)

- Used to create a precise voltage divider for setting Vref to exactly 100 mV.
- Multi-turn trimmer recommended for fine adjustment.

Passive Components (SMD 0805)

Resistors and capacitors were selected in SMD 0805 package for the PCB design because:

- Compact size allows smaller PCB footprint.
- Widely available and easy to hand-solder with basic tools.
- Sufficient power rating (1/8W) for this low-power circuit.

Components Used in Breadboard Prototype

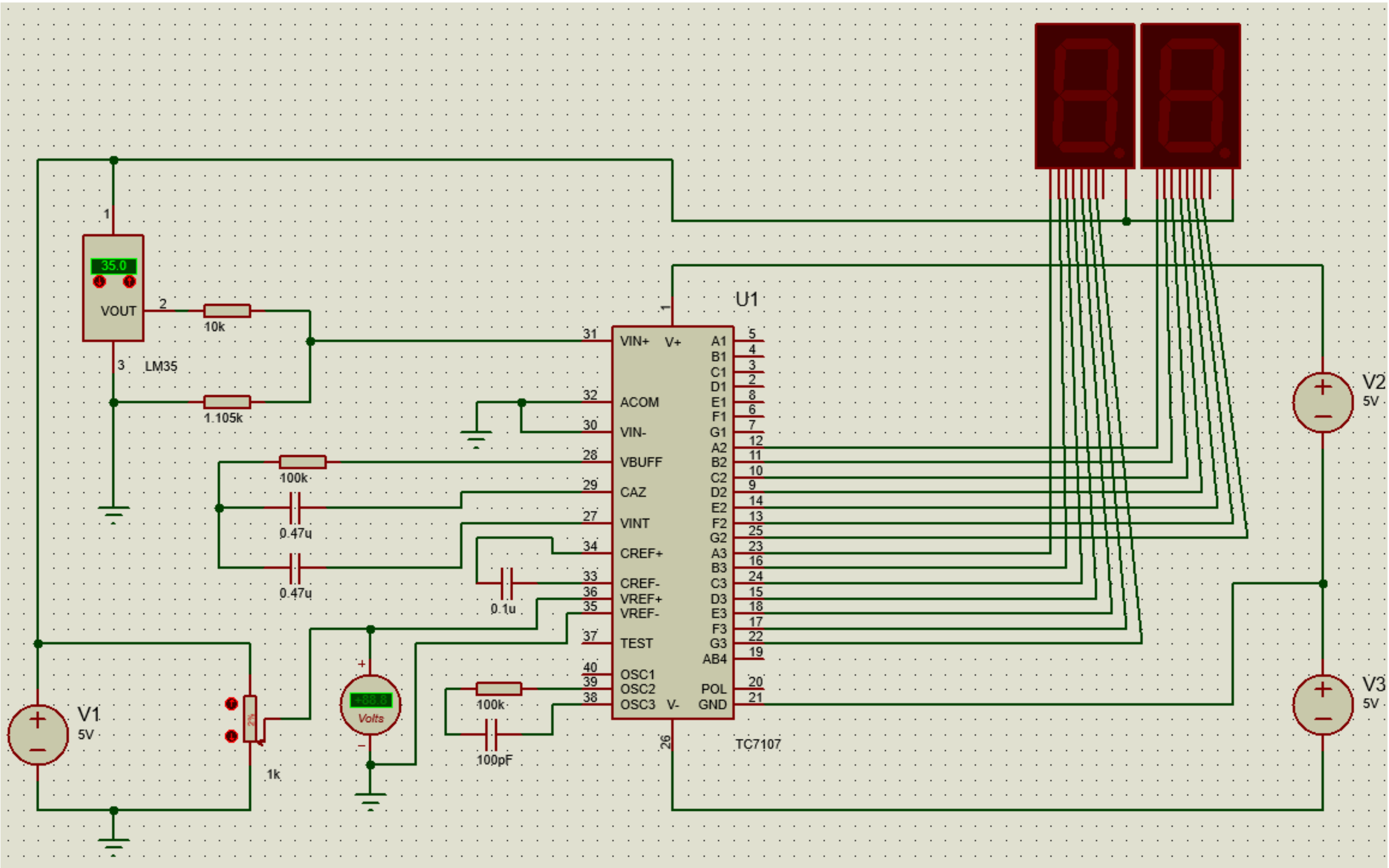
For the breadboard prototype, through-hole equivalents of resistors and capacitors were used instead of SMD 0805, as breadboards cannot accommodate SMD packages. Otherwise, the component selection remained identical to the PCB design.

Results & Testing

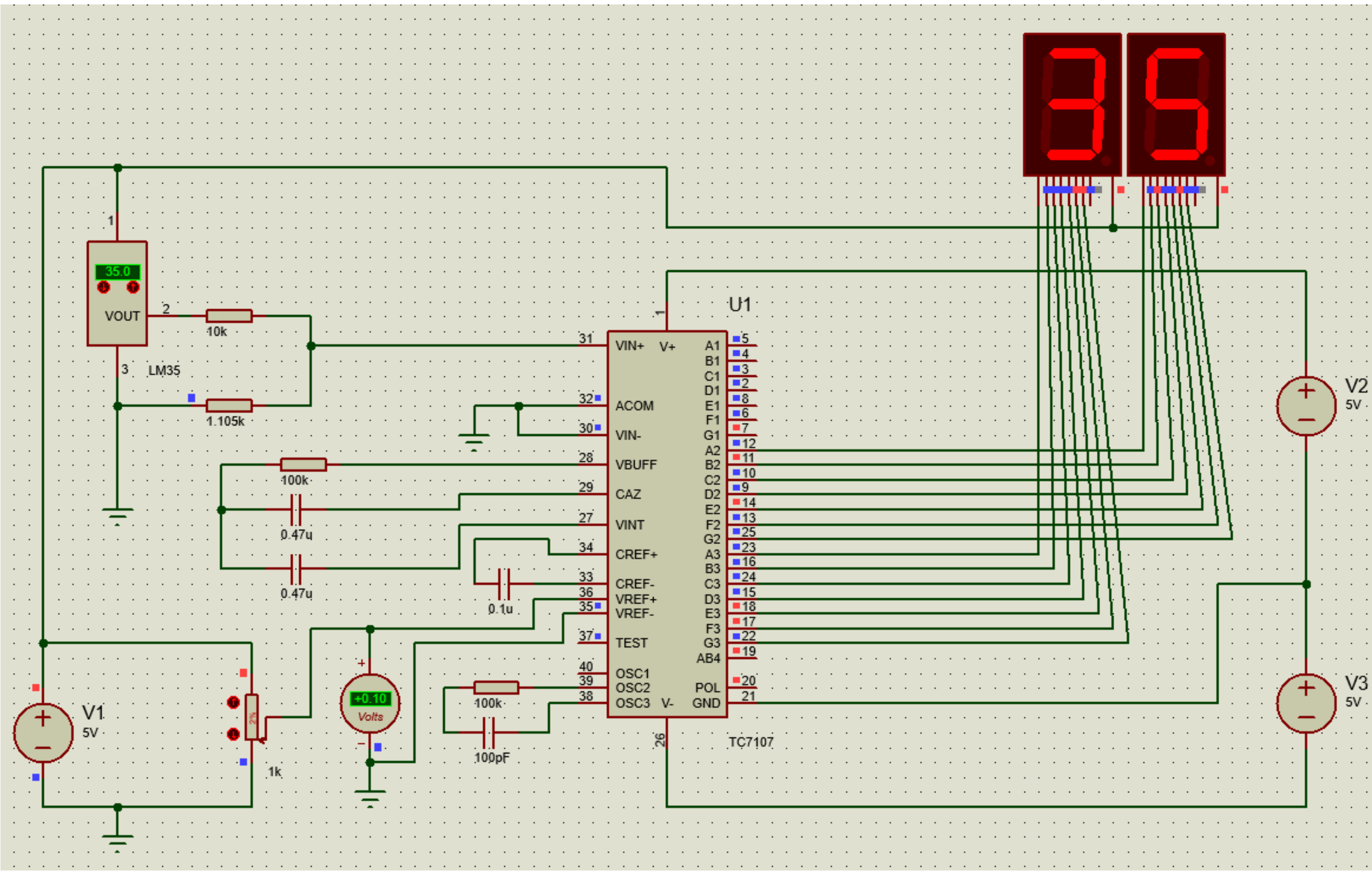
This section presents the results from all three phases of the project: successful Proteus simulation, completed PCB design, and the failed breadboard prototype showing the specific "66" issue.

Proteus Simulation Results

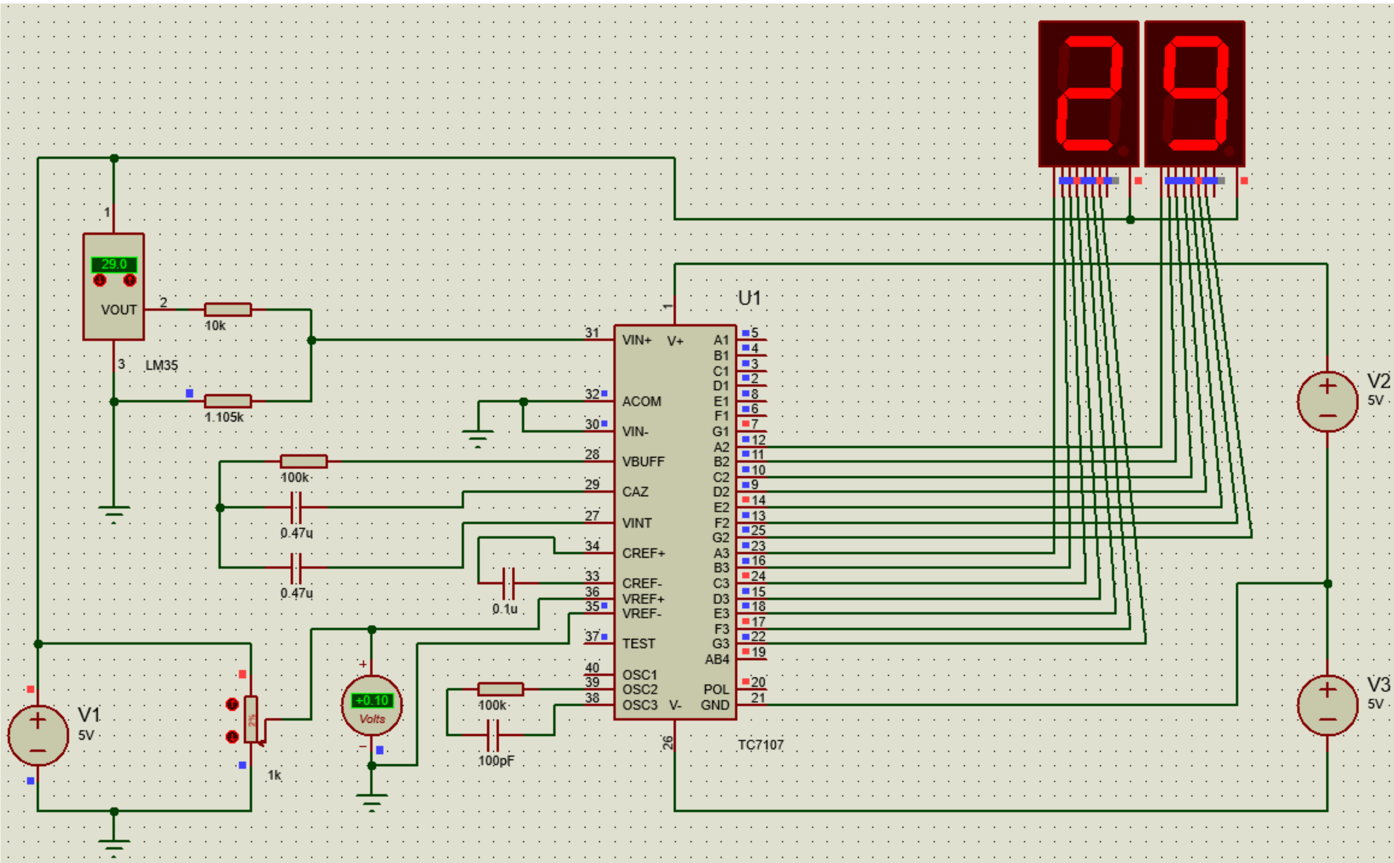
The circuit was simulated in Proteus across the full temperature range. The LM35 output was varied from 0 mV to 990 mV, corresponding to 0°C to 99°C. The ICL7107 reference voltage (Vref) was adjusted to exactly 100 mV using a potentiometer.



Schematic Diagram



Output 01



Output 02

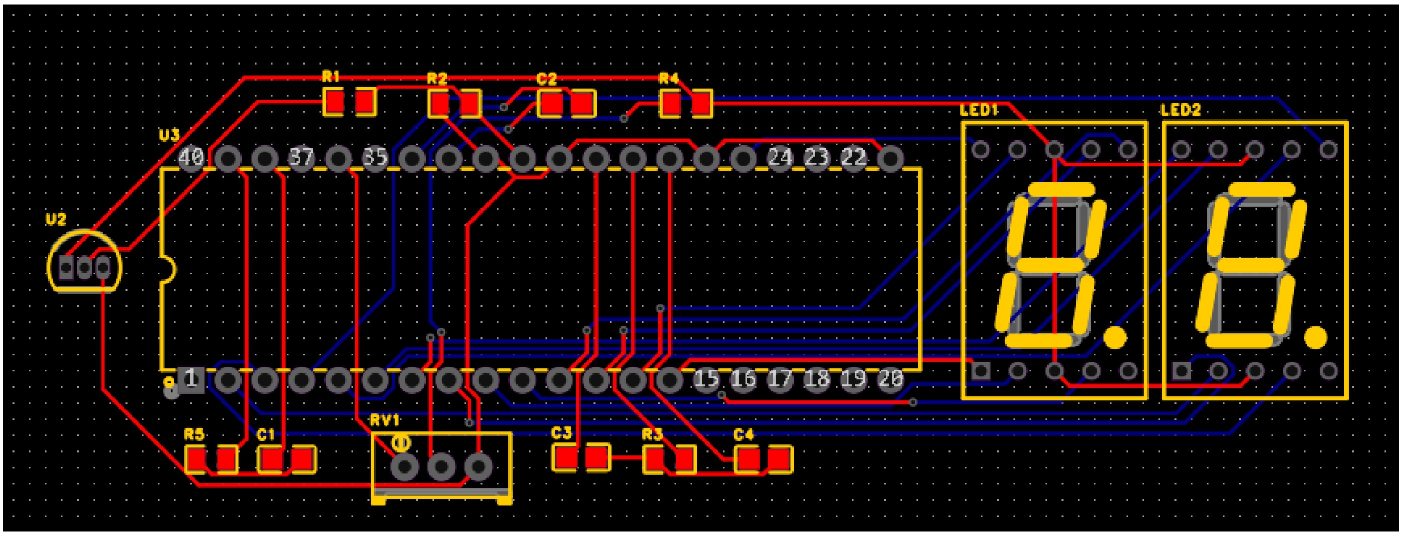
Observation: The display accurately tracked the temperature across the entire 0°C to 99°C range with no errors, jitter, or missing segments.

Conclusion from simulation: The voltage scaling, reference adjustment, and display driving logic are all correct. The ICL7107, when properly configured in software, accurately converts the LM35 analog output to the corresponding temperature on two 7-segment displays.

Phase II: EasyEDA PCB Design

The PCB layout was successfully completed in EasyEDA Pro.

Requirement	Status
LM35 in through-hole (TO-92)	Implemented
ICL7107 in DIP-40	Implemented
Resistors and capacitors in SMD 0805	Implemented
Star grounding for analog signals	Implemented
Two 7-segment displays horizontally aligned	Implemented
Design Rule Check (DRC) passed	Completed
Gerber files generated	Completed



PCB Design

Visual inspection: The PCB layout shows clean routing, proper component placement, and clear separation of analog and digital ground paths meeting at a single star point.

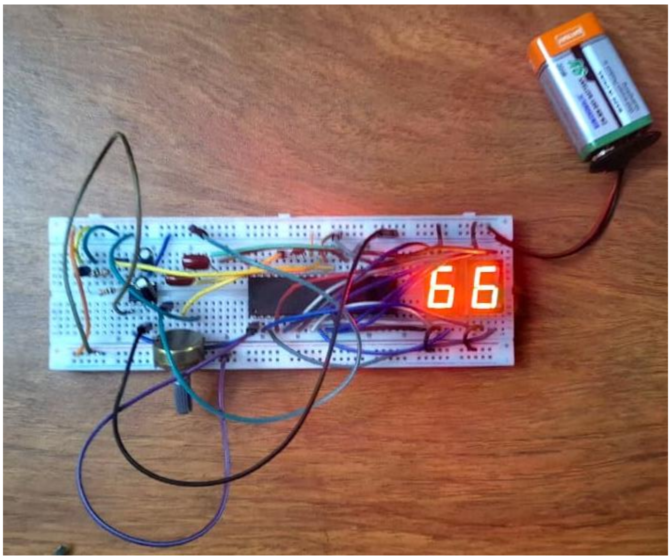
Phase III: Breadboard Prototype Results

The identical circuit was constructed on a breadboard using the same schematic as the simulation and PCB design.

Observed Behavior:

Upon applying power, both 7-segment displays illuminated but showed a fixed reading.

Test Condition	Expected Display	Actual Display
LM35 at room temperature (~25°C, ~250 mV)	25	66
LM35 heated with fingers (~35°C, ~350 mV)	35	66
LM35 cooled with ice (~5°C, ~50 mV)	05	66
External 0 mV input (short to ground)	00	66
External 250 mV input (from power supply)	25	66
External 500 mV input	50	66
External 990 mV input	99	66
Vref adjusted from 50 mV to 200 mV	Varies	66 (no change)



Breadboard Prototype

Summary of Results

Phase	Outcome	Key Observation
Proteus Simulation	Success	Accurate display from 0°C to 99°C
EasyEDA PCB Design	Success	Gerber files ready, star grounding implemented
Breadboard Prototype	Failure	Fixed "66" display regardless of input

Discussion

This section analyzes the discrepancy between the successful simulation and the failed breadboard prototype, with particular focus on the specific "66" failure mode. Probable causes are identified, ranked by likelihood, and lessons learned are discussed.

Analysis of the "66" Failure Mode

A fixed "66" display is highly informative. Unlike random noise or flickering digits, a stable "66" indicates a deterministic failure rather than an intermittent connection or simple wiring error. The pattern "6" corresponds to all 7-segment display segments illuminated except segment 'g' (the middle horizontal segment).

What "66" tells us:

- Both digits are being driven with the same pattern.
- The segment outputs are not completely dead (some signals are reaching the display).
- The ADC is not updating the display based on the input voltage.

Probable Causes (Ranked by Likelihood)

Rank	Possible Cause	Explanation	Why "66" Occurs
1	Missing oscillator circuit	The ICL7107 requires an external RC network between pins 39 (OSC1) and 40 (OSC2) to generate its internal clock. Without this, the ADC does not run, and the display latches into a default or undefined state.	The chip powers up but never updates the display registers, leaving a default pattern (often all segments except 'g')
2	Missing negative power supply (-5V)	The ICL7107 requires a dual supply of ±5V for proper operation. If V- (pin 26) is connected to ground instead of -5V, the internal analog circuitry does not function correctly.	The ADC cannot process the input voltage, and the display driver may default to "66"
3	Incorrect reference voltage connection	If Vref+ and Vref- are not properly connected or floating, the ADC has no reference to measure against.	The ADC outputs an arbitrary or mid-scale value that happens to decode to "66"
4	Floating analog inputs	If IN+ (pin 31) or IN- (pin 30) are not properly referenced to analog ground, the ADC may read a constant mid-scale value.	A mid-scale input (~500 mV) would normally display "50", but with other faults could appear as "66"
5	Damaged or counterfeit ICL7107	Some DIP-40 ICL7107 chips from online suppliers are counterfeit or static-damaged.	A fixed "66" is a known symptom of a faulty chip in some student project forums
6	Incorrect display	The ICL7107 is designed for common anode displays. If common cathode displays are used	A corrupted pattern could accidentally produce "6" on both digits

	common anode wiring	or wiring is incorrect, segment patterns can be corrupted.	

Additional Observations

Positive findings despite failure:

- The LM35 sensor was correctly wired and producing accurate output voltages.
- The 7-segment displays were receiving power and showing a stable pattern (not dead).
- The simulation confirmed that the schematic design is logically correct.
- The PCB layout was completed correctly and is ready for fabrication.

What did NOT cause the failure:

- LM35 sensor (confirmed working with multimeter)
- Basic wiring of display segments (some segments illuminated)
- Reference voltage potentiometer (could be adjusted, though no effect on display)

Lessons Learned

1. **Read the datasheet thoroughly** – The ICL7107 datasheet clearly specifies the need for an external oscillator and ±5V supply. These requirements were overlooked.
2. **Simulation is not a substitute for hardware knowledge** – A simulation can pass perfectly while hardware fails due to missing components that the simulation does not require.
3. **Test systematically** – Isolating the problem by testing with external voltage sources (instead of the LM35) helped confirm that the issue was in the ICL7107 stage, not the sensor.
4. **Start simple** – Testing with a single 7-segment display before adding the second would have simplified debugging.
5. **Document failures** – The "66" pattern is a valuable diagnostic clue that points specifically to oscillator and power supply issues.

Conclusion

This project aimed to design and build a 2-digit digital thermometer using an LM35 temperature sensor and an ICL7107 ADC with integrated 7-segment display drivers, capable of measuring temperatures from 0°C to 99°C.

The work was completed in three phases. First, the circuit was successfully simulated in Proteus, where the reference voltage was set to exactly 100 mV, and the display accurately tracked the LM35 output from 0°C to 99°C. Second, a PCB layout was completed in EasyEDA Pro, incorporating star grounding for analog signal integrity and horizontal alignment of the two 7-segment displays. Gerber files were generated and are ready for fabrication.

Third, a breadboard prototype was constructed to validate the hardware before PCB assembly. This phase was unsuccessful. The display consistently showed a fixed **"66"** regardless of input voltage, temperature applied to the LM35, or adjustments to the reference voltage potentiometer. The LM35 sensor was verified to be functioning correctly with a multimeter.

Analysis of the failure identified the most probable root causes as:

1. **Missing oscillator circuit** – No resistor or capacitor was connected to pins 39 and 40 of the ICL7107. Without an external clock, the ADC does not run, and the display latches into a default pattern.
2. **Missing negative power supply (-5V)** – The ICL7107 requires a ±5V dual supply, but only a single +5V supply was provided. The datasheet specifies that V- (pin 26) must be connected to -5V, not ground.

These requirements were overlooked because the Proteus simulation environment automatically provides ideal internal clocking and power supply conditions, masking the need for external oscillator components and dual supply voltages in hardware.

A structured future correction plan has been developed, prioritizing the addition of the oscillator circuit (100 pF capacitor and 100 kΩ resistor) and a proper ±5V power supply using either an ICL7660 charge pump or 7805/7905 regulators. Once these corrections are implemented on the breadboard, the verified circuit will be transferred to the already-designed PCB for final assembly.

Despite the hardware failure, the project successfully demonstrated the simulation and PCB design phases. The experience reinforced a critical engineering lesson: **simulation validates logic but does not guarantee hardware functionality**. Attention to datasheet requirements, power supply design, and oscillator circuits is essential when transitioning from simulation to physical implementation.

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